

Understanding Different Magnetic Materials

To understand magnetic materials, it is useful to begin with the idea of a **magnetic dipole moment**.

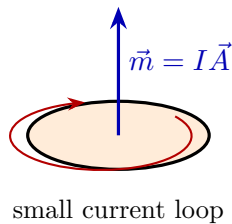
A small current loop of current I and area vector $\vec{A}$ has magnetic dipole moment

$$\vec{m} = I\vec{A} \quad (1)$$

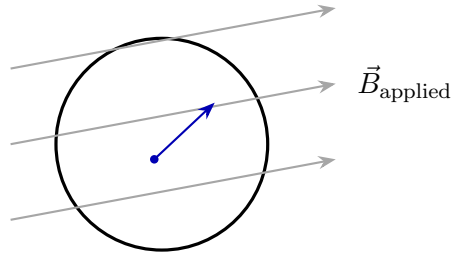
whose direction is given by the right-hand rule.

In atoms, magnetic dipole moments arise mainly due to

- orbital motion of electrons,
- electron spin,
- and, in some cases, weak contributions from the nucleus.



small current loop



material made of many atomic dipoles

Figure 1: A magnetic dipole moment can be modeled by a small current loop. Magnetic materials respond because atoms may have magnetic dipole moments.

The key question is: *how do these atomic moments behave collectively inside a material?* That gives rise to the broad classes of magnetic materials.

Case 1: Diamagnetic materials

In many atoms, the various electronic contributions cancel so that the atom has *no permanent net magnetic dipole moment* in the absence of an external field.

However, when an external magnetic field is applied, the electron motion is slightly modified. This induces a weak magnetic dipole moment that *opposes* the applied field.

So diamagnetism is characterized by:

- no permanent atomic magnetic moment,



- only an *induced* moment in the presence of applied field,
- the induced moment opposes the applied field,
- hence the material is weakly repelled by strong magnetic fields.

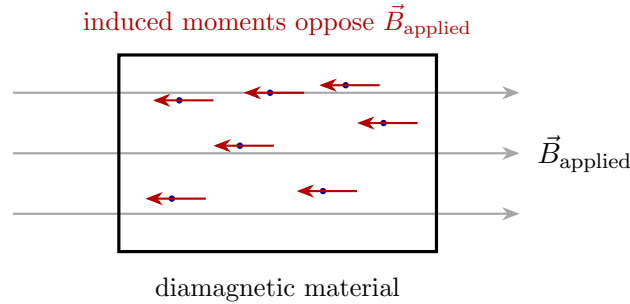


Figure 2: In a diamagnetic material, there is no permanent atomic moment; the applied field induces small moments opposite to the field.

Examples include copper, silver, bismuth, and water (very weakly diamagnetic in bulk).

Case 2: Paramagnetic materials

Some atoms do possess a *permanent* magnetic dipole moment. But in the absence of an external field, thermal agitation makes these moments point in random directions. Therefore, the material as a whole has nearly zero net magnetic moment.

When an external magnetic field is applied, some of these moments tend to align with the field. The alignment is only partial, because thermal motion still tends to randomize them.

So paramagnetism is characterized by:

- permanent atomic magnetic moments,
- random orientation without applied field,
- partial alignment with applied field,
- weak attraction toward strong magnetic fields.

Examples include aluminum, platinum, and oxygen gas.

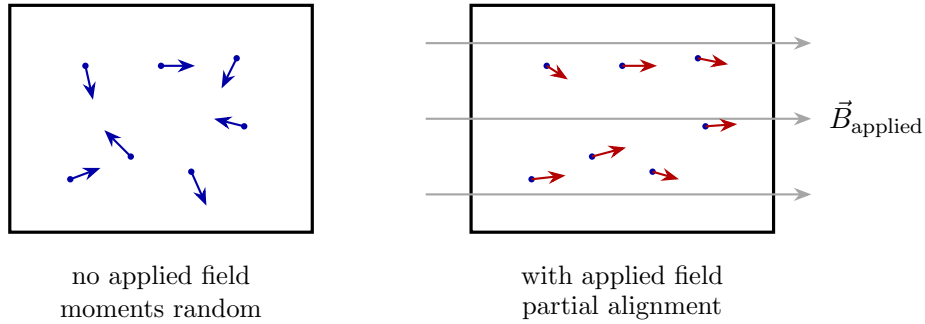


Figure 3: Paramagnetic material: permanent atomic moments are random without applied field, but tend to align weakly with the applied field.

Case 3: Ferromagnetic materials

In some materials, the magnetic interaction between neighboring atoms is so strong that many nearby atomic moments align in the same direction even without an external field. These aligned regions are called **domains**.

Inside each domain, the net magnetic moment is large. However, different domains may initially point in different directions, so the material as a whole may still show little or no net magnetization.

When an external field is applied, domains favorably oriented with the field grow at the expense of others, and the material becomes strongly magnetized.

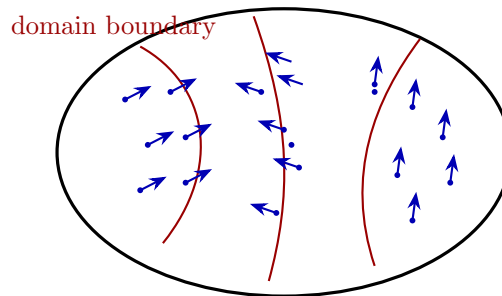


Figure 4: Ferromagnetic material: atomic moments align strongly within domains. Different domains may initially point in different directions.

Ferromagnets respond very strongly to magnetic fields. Moreover, after removing the external field, some ferromagnetic materials may retain part of their magnetization. This is the basis of permanent magnets.

Examples include iron, cobalt, nickel, and many ferrites.

Comparison of the three cases

Comparison of Magnetic Materials

1. **Diamagnetic materials:** atoms have no permanent net magnetic moment; the applied field induces a weak moment opposite to the field.
2. **Paramagnetic materials:** atoms have permanent magnetic moments, but they are random without applied field; the field causes weak partial alignment.
3. **Ferromagnetic materials:** strong interaction aligns neighboring atomic moments into domains; the material can become strongly magnetized and may retain magnetization.

A correction to a common intuition: electric materials also have analogs

It is tempting to say that electric materials do not show such variety, but that is not fully correct.

There are electrical analogs too:

- ordinary dielectrics,
- polar dielectrics,
- ferroelectrics.

However, in basic electromagnetics we often emphasize magnetic materials more because magnetic dipoles arising from electron spin and orbital motion naturally lead to dia-, para-, and ferro-magnetic behavior.

For electric materials, bulk matter is usually nearly charge neutral, and strong Coulomb forces tend to suppress large static charge imbalance. So the electric analog is present, but often discussed under dielectric polarization rather than in exactly the same language as magnetic materials.

Side Remark

Water molecules, for example, have a permanent *electric dipole moment*, but in liquid water the molecular orientations are usually random in the absence of an applied field. So the *bulk* material may show little net polarization unless an external field is applied.

Magnetic Field in Materials

To understand magnetic materials more deeply, we should first understand how a **magnetic dipole** (or equivalently a small current loop) responds to an externally applied magnetic field.

A current loop in a uniform magnetic field

Consider a rectangular current loop placed in a uniform magnetic field $\vec{B}$. A magnetic dipole moment is associated with the loop:

$$\vec{m} = I\vec{A} \quad (2)$$

where $\vec{A}$ is the area vector normal to the plane of the loop, given by the right-hand rule.

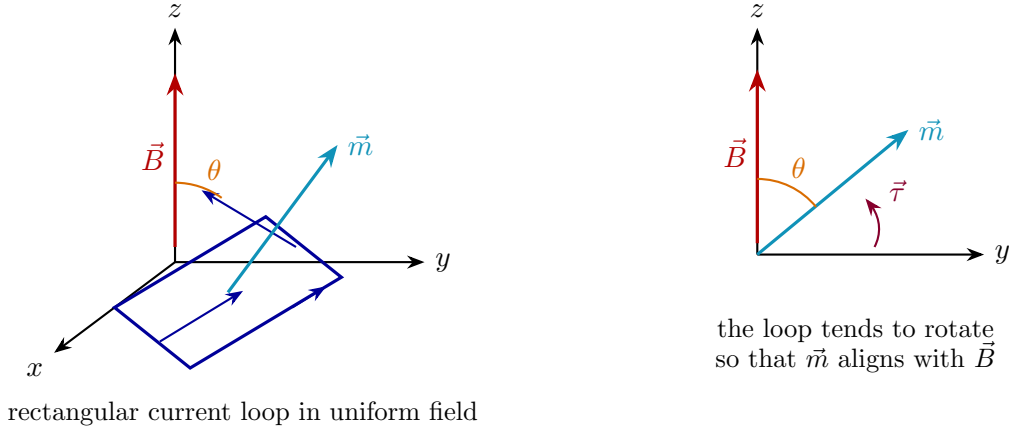


Figure 5: A current loop in a uniform magnetic field. The net force is zero, but a torque acts to align the magnetic moment with the applied field.

The magnetic forces on opposite sides of the loop are equal and opposite, so the **net force is zero**. However, they form a couple and therefore produce a **torque**.

The result is

$$\vec{\tau} = \vec{m} \times \vec{B} \quad (3)$$

and hence the magnitude is

$$\tau = mB \sin \theta. \quad (4)$$

So the effect of the applied field is to rotate the dipole so that $\vec{m}$ tends to align with $\vec{B}$.

Key Result

A magnetic dipole in a uniform magnetic field experiences

$$\vec{\tau} = \vec{m} \times \vec{B}.$$

Thus, even when the net force is zero, the dipole tends to rotate and align with the external field.

What happens inside a material?

Now imagine a material made up of a very large number of atoms, each carrying a tiny magnetic dipole moment.

Before applying any external magnetic field, these dipole moments are often randomly oriented. Hence the *net magnetic moment* of the specimen may be nearly zero.

When an external magnetic field is applied, each atomic dipole experiences a torque and tends to align with the field.

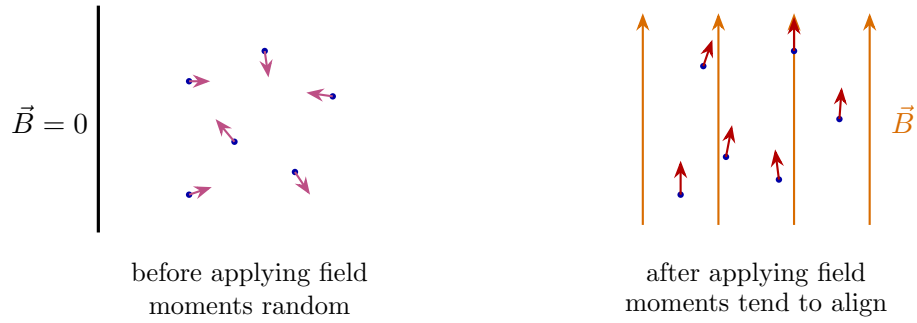


Figure 6: Before applying an external field, atomic magnetic moments may be random. Under an applied field, the moments tend to align, producing a nonzero net magnetic effect.

Thus the basic physical idea is:

- before applying $\vec{B}$, the atomic moments may cancel on average,
- after applying $\vec{B}$, they partially align,
- therefore the material develops a **net magnetic moment**.

From microscopic moments to macroscopic magnetization

At this point, it is better to introduce a more careful macroscopic quantity.

Strictly speaking, $\vec{m}$ is the dipole moment of an individual atom or molecule, so the more proper macroscopic quantity is the **magnetization**

$$\vec{M} = \frac{\text{magnetic dipole moment}}{\text{unit volume}}. \quad (5)$$

Therefore, the total magnetic dipole moment of the specimen is

$$\vec{m}_{\text{net}} = \int_V \vec{M} dv. \quad (6)$$

So we may summarize:

- before applying the field,

$$\int_V \vec{M} dv \approx 0,$$

- after applying the field,

$$\int_V \vec{M} dv \neq 0.$$

That is, the material becomes magnetized.

Induced field inside the material

The aligned magnetic dipoles inside the material produce their own magnetic field. This contribution is often called the **magnetization field** or **induced magnetic field**.

So the total magnetic field inside matter is not just the externally applied field; rather, it contains two contributions:

$$\boxed{\vec{B}_{\text{total}} = \vec{B}_{\text{applied}} + \vec{B}_{\text{induced}}.} \quad (7)$$

For weak linear magnetic materials, the induced effect is proportional to the applied field (more precisely, proportional to $\vec{H}$):

$$\vec{M} = \chi_m \vec{H}. \quad (8)$$

Then

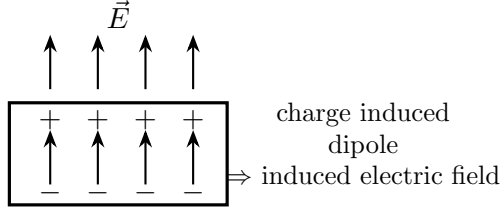
$$\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0(1 + \chi_m)\vec{H} = \mu\vec{H}. \quad (9)$$

Thus:

- in **diamagnetic** materials, the induced response opposes the applied field,
- in **paramagnetic** materials, it supports the applied field weakly,
- in **ferromagnetic** materials, the response can be very strong and nonlinear.

Side Note: Analogy with Electrostatics in Matter

See the analogy with electrostatics:



$$\vec{E}_{\text{ind}} \propto \vec{E}$$

$$\epsilon_0 \nabla \cdot \vec{E} = \rho_f + \rho_b$$

$$\epsilon_0 \nabla \cdot \vec{E}_{\text{ind}} = \rho_b$$

$$\epsilon_0 \nabla \cdot (\vec{E} - \vec{E}_{\text{ind}}) = \rho_f$$

Define

$$\vec{D} = \epsilon_0 (\vec{E} - \vec{E}_{\text{ind}})$$

Hence,

$$\nabla \cdot \vec{D} = \rho_f$$

To deal with electric field in matter

Similarly to electrostatics, do the same analysis in magnetic field in material:

$$\nabla \times \frac{\vec{B}}{\mu_0} = \vec{J}_{\text{total}} = \vec{J}_f + \vec{J}_b$$

$$\nabla \times \frac{\vec{B}_{\text{ind}}}{\mu_0} = \vec{J}_b$$

$$\nabla \times \left(\frac{\vec{B} - \vec{B}_{\text{ind}}}{\mu_0} \right) = \vec{J}_f$$

Define

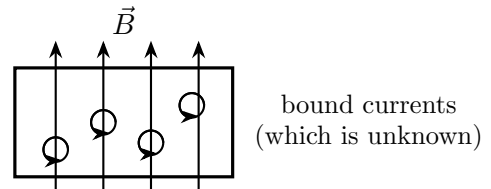
$$\vec{H} = \frac{\vec{B} - \vec{B}_{\text{ind}}}{\mu_0}$$

Hence,

$$\nabla \times \vec{H} = \vec{J}_f$$

or in integral form,

$$\oint \vec{H} \cdot d\vec{l} = I_{\text{enc, free}}$$



We can use $\vec{H}$ -field forgetting about bound currents (which is unknown), but considering only free currents.

To deal with magnetic field in matter

In electromagnetism, $\vec{E}$ and $\vec{B}$ are primary fields, while $\vec{D}$ and $\vec{H}$ are derived fields.

Connecting with the previous classification of materials

This picture now explains the three cases discussed earlier:

1. **Diamagnetic materials:** no permanent atomic moment; an applied field induces weak moments opposing the field.
2. **Paramagnetic materials:** permanent atomic moments exist, but are randomly oriented until the applied field partially aligns them.
3. **Ferromagnetic materials:** strong interaction between neighboring moments causes cooperative alignment, giving a much larger magnetization.

Boxed Summary

1. A current loop or magnetic dipole in a uniform magnetic field experiences no net force, but it does experience torque:

$$\vec{\tau} = \vec{m} \times \vec{B}.$$

2. Therefore, a magnetic dipole tends to align with the applied field.
3. In a material, many atomic dipole moments are present. Without applied field, their average may be zero.
4. When an external field is applied, these dipoles tend to align, producing a nonzero **magnetization** $\vec{M}$.
5. The material then produces its own magnetic contribution, so the total field inside matter is modified.
6. For weak linear materials,

$$\vec{M} = \chi_m \vec{H}, \quad \vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu \vec{H}.$$

Scalar Magnetic Potential

We now discuss an idea that is very useful in magnetostatics, especially in regions where there is **no free current**. This is the concept of the **scalar magnetic potential**.

1. Starting point

Recall the two important magnetostatic relations:



$$\nabla \cdot \vec{B} = 0 \quad (10)$$

which is always true, even in the presence of magnetic material, and

$$\nabla \times \vec{H} = \vec{J}_f \quad (11)$$

which tells us that the curl of $\vec{H}$ is determined by the **free current density**.

Therefore, in a region where

$$\vec{J}_f = 0, \quad (12)$$

we have

$$\nabla \times \vec{H} = 0. \quad (13)$$

This is the key condition for defining a scalar magnetic potential.

Key Idea

The scalar magnetic potential can be introduced only in a region where

$$\nabla \times \vec{H} = 0,$$

that is, in a **free-current-free region**.

2. Analogy with electrostatics

In electrostatics,

$$\nabla \times \vec{E} = 0 \quad (14)$$

everywhere, and therefore the line integral of $\vec{E}$ is path independent:

$$\int_1^2 \vec{E} \cdot d\vec{l} = f(\vec{r}_1, \vec{r}_2). \quad (15)$$

Hence we define the electric scalar potential ϕ through

$$\vec{E} = -\nabla\phi. \quad (16)$$

In magnetostatics, a similar idea works for $\vec{H}$, but only in regions where $\vec{J}_f = 0$.

If

$$\nabla \times \vec{H} = 0, \quad (17)$$

then

$$\oint_C \vec{H} \cdot d\vec{l} = 0 \quad (18)$$

for any closed path C lying entirely in that region. Hence the line integral of $\vec{H}$ between two points becomes path independent, and we may define

$$\boxed{\vec{H} = -\nabla\phi_m} \quad (19)$$

where ϕ_m is called the **scalar magnetic potential**.

Electrostatic Analogy

$$\begin{aligned} \nabla \times \vec{E} = 0 &\implies \vec{E} = -\nabla\phi \\ \nabla \times \vec{H} = 0 &\implies \vec{H} = -\nabla\phi_m \end{aligned}$$

But note carefully: the second relation is valid only in a **free-current-free region**.

3. Geometric picture

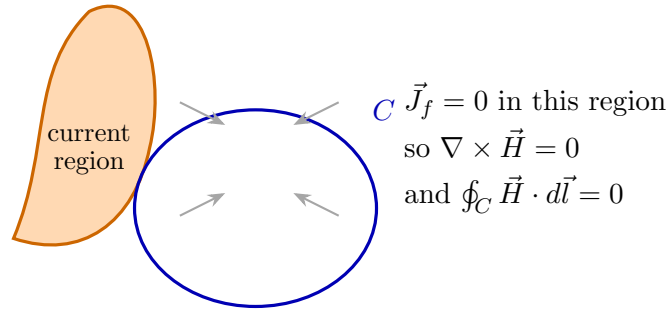


Figure 7: In a region containing no free current, $\nabla \times \vec{H} = 0$. Hence the line integral of $\vec{H}$ around any closed path lying entirely in that region is zero.

So, if two points $\vec{r}_1$ and $\vec{r}_2$ lie in such a region, then

$$\int_{\vec{r}_1}^{\vec{r}_2} \vec{H} \cdot d\vec{l} \quad (20)$$

depends only on the endpoints and not on the path, provided the path remains entirely in the current-free region.

Thus,

$$\boxed{\phi_m(\vec{r}_2) - \phi_m(\vec{r}_1) = - \int_{\vec{r}_1}^{\vec{r}_2} \vec{H} \cdot d\vec{l}.} \quad (21)$$

4. Important limitation: not valid around current-carrying conductors

The above statement must be used with care.

If a path encloses a free current, then from Ampère's law,

$$\oint_C \vec{H} \cdot d\vec{l} = I_{\text{enc, free}} \neq 0. \quad (22)$$

Hence the line integral is no longer zero, and the scalar magnetic potential cannot be globally single-valued in such a region.

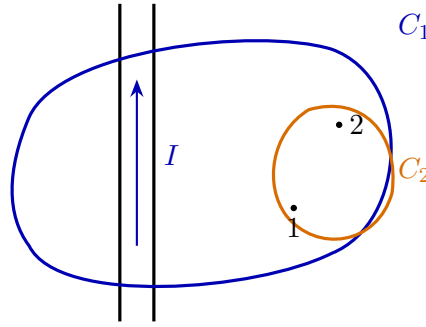


Figure 8: The orange path C_2 does not enclose the current, while the blue path C_1 wraps around the current-carrying conductor.

For a path such as C_2 , which does not enclose the current,

$$\int_1^2 \vec{H} \cdot d\vec{l} \quad (23)$$

is path independent within that current-free simply connected region.

But for a path such as C_1 , which wraps around the conductor,

$$\oint_{C_1} \vec{H} \cdot d\vec{l} = I. \quad (24)$$

More generally, if a path winds around the conductor n times,

$$\oint_C \vec{H} \cdot d\vec{l} = nI. \quad (25)$$

So the scalar magnetic potential is valid only in a region where the path does not link free current.

Important Remark

The scalar magnetic potential is not globally defined in regions that contain linked free currents. It is valid only in **free-current-free simply connected regions**.

5. Example: use in magnetic circuits

This idea is very useful in magnetic circuits, transformers, and high-permeability cores, where one often analyzes current-free regions of the magnetic path.

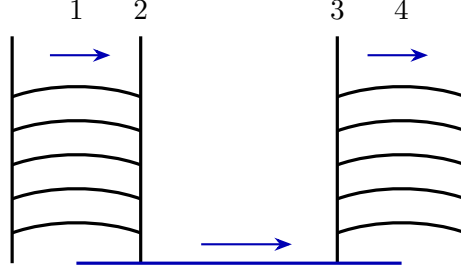


Figure 9: In magnetic circuits and transformer-type structures, one often analyzes segments of the magnetic path using scalar-potential ideas in current-free regions.

For example, along a current-free segment,

$$\int_1^2 \vec{H} \cdot d\vec{l} = f(1, 2), \quad \int_3^4 \vec{H} \cdot d\vec{l} = f(3, 4), \quad (26)$$

which reduces the problem to a scalar-potential picture in those regions.

Relation with $\vec{B}$

Since

$$\vec{H} = -\nabla\phi_m, \quad (27)$$

in a linear homogeneous medium where

$$\vec{B} = \mu\vec{H}, \quad (28)$$

we get

$$\vec{B} = -\mu\nabla\phi_m. \quad (29)$$

Then using

$$\nabla \cdot \vec{B} = 0, \quad (30)$$

we obtain

$$\nabla \cdot (\mu\nabla\phi_m) = 0. \quad (31)$$

If μ is constant in the region, this reduces to

$$\boxed{\nabla^2\phi_m = 0.} \quad (32)$$

So in a homogeneous current-free region, the scalar magnetic potential satisfies Laplace's equation.

Boxed Summary

1. In magnetostatics,

$$\nabla \times \vec{H} = \vec{J}_f.$$

2. Therefore, in a region where $\vec{J}_f = 0$,

$$\nabla \times \vec{H} = 0.$$

3. In such a free-current-free simply connected region, the line integral of $\vec{H}$ is path independent, so we may define the scalar magnetic potential:

$$\vec{H} = -\nabla \phi_m.$$

4. This is analogous to electrostatics, where

$$\vec{E} = -\nabla \phi$$

because $\nabla \times \vec{E} = 0$.

5. If a path encloses free current, then

$$\oint \vec{H} \cdot d\vec{l} = I_{\text{enc,free}} \neq 0,$$

so a single-valued scalar magnetic potential cannot be used globally.

6. In a homogeneous current-free region,

$$\nabla^2 \phi_m = 0.$$